

Fundamentals of Electric Circuits

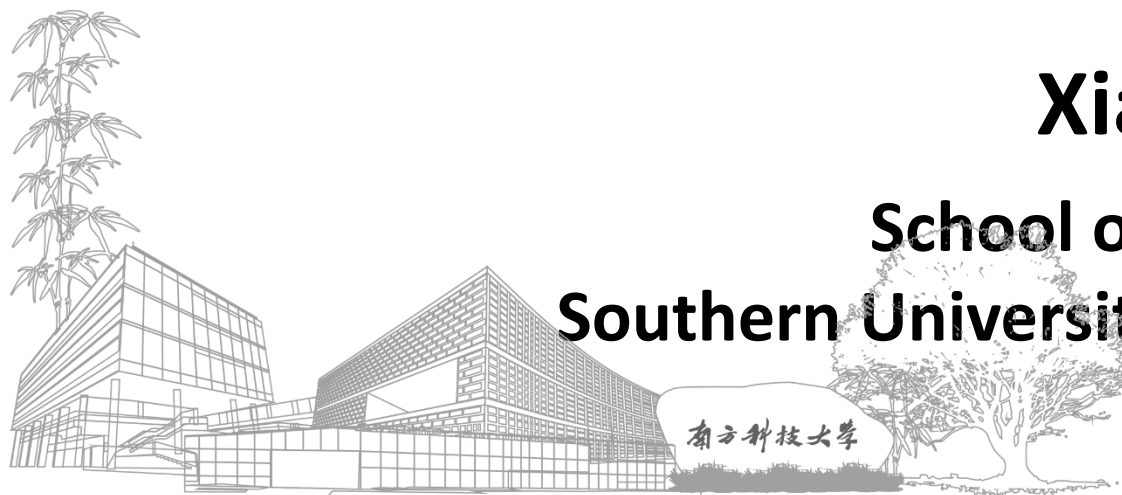
电路基础

Lecture7a: Differential Equation Review

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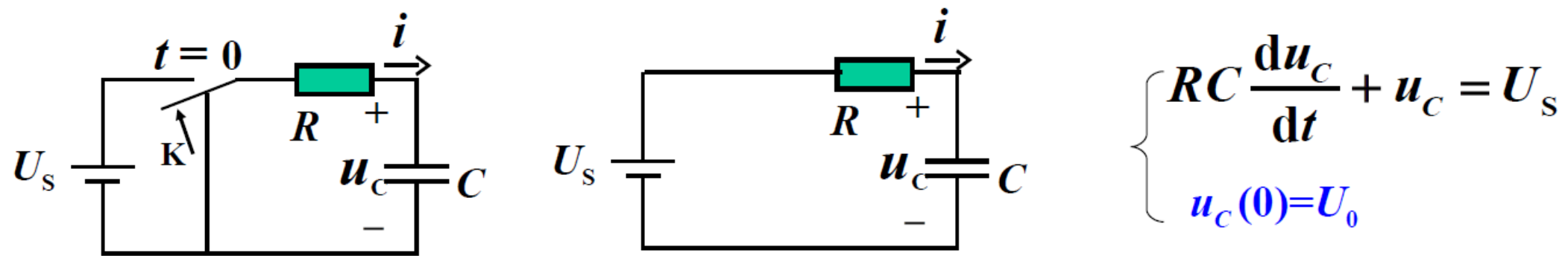


Outline

- **Introduction**
- **First-Order Differential Equation**
- **Second-Order Differential Equation**

Introduction (I)

- Capacitors and inductors have voltage-current (or current-voltage) relationships described by differentials, i.e., $i_C = C \cdot dv_C/dt$, $v_L = L \cdot di_L/dt$
- The analysis of circuits that contains the storage elements involves finding solutions, i.e., the currents and voltages as functions of time, for **differential equations (微分方程)**, i.e., equations that relate functions with their **derivatives (导数)**
- We will review some basics regarding differential equations. We will put emphasis on how to solve the **first-order** and **second-order linear differential equation with constant coefficients (常系数线性微分方程)**



Introduction (II)

- We will review some basics regarding differential equations. We will put emphasis on how to solve the **first-order** and **second-order linear differential equations with constant coefficients** (常系数线性微分方程)
- Linear means that the variable in an equation appears only with a **power of one** (一次方). So 'x' is linear, but 'x²' or cos(x) is non-linear
- In a differential equation, when the variables and their derivatives are only multiplied by constants, then the equation is linear. The variables and their derivatives must always appear as a simple first power. For example, equation (R1.1) is linear, while (R1.2) or (R1.3) is non-linear:

$$y'' + y' + y = 0 \text{ (R1.1); } (y'')^2 + y' + y = 0 \text{ (R1.2); } y'' + \cos(y') + y = 0 \text{ (R1.3)}$$

- However, an exception is made for the time variable t (i.e., the variable that we are differentiating by), the following equations are all linear in y

$$y' + 2y = \cos(t) \text{ (R1.4); } y'' + 2y' + y = \sin(t) \text{ (R1.5); } y' + t^2 y = 0 \text{ (R1.6)}$$

Introduction (III)

- Moreover, we will be dealing with **linear and time-invariant (线性时不变)** circuits (and systems), so we focus on linear differential equations whose **variables and their derivatives are multiplied by time-invariant constants**, such as those in (R1.4) and (R1.5), rather than the one in (R1.6)

$$y' + 2y = \cos(t) \quad (\text{R1.4}); \quad y'' + 2y' + y = \sin(t) \quad (\text{R1.5}); \quad y' + t^2 y = 0 \quad (\text{R1.6})$$

- The order of a differential equation (微分方程阶数) is equal to the highest derivative in the equation. Equation (R1.4) is first-order while (R1.5) are second-order
- A linear differential equation is **homogeneous (齐次)** if $g(t)=0$, and nonhomogeneous if $g(t) \neq 0$. So, equation (R1.1) is homogeneous while (R1.4) is **non-homogeneous**

$$k_N y^{(N)} + k_{N-1} y^{(N-1)} + \dots + k_2 y'' + k_1 y' + k_0 y = g(t)$$

- The nonhomogeneous part $g(t)$ usually corresponds to a **forcing term (强迫项)** in the physical model. For example, an independent source in a circuit

Mathematical Formulas

C.4 Derivatives

If $U = U(x)$, $V = V(x)$, and $a = \text{constant}$

$$\frac{d}{dx}(aU) = a \frac{dU}{dx} \qquad \frac{d}{dx}\left(\frac{U}{V}\right) = \frac{V \frac{dU}{dx} - U \frac{dV}{dx}}{V^2}$$

$$\frac{d}{dx}(UV) = U \frac{dV}{dx} + V \frac{dU}{dx} \qquad \frac{d}{dx}(aU^n) = naU^{n-1} \frac{dU}{dx}$$

$$\frac{d}{dx}(a^U) = a^U \ln a \frac{dU}{dx} \qquad \frac{d}{dx}(\sin U) = \cos U \frac{dU}{dx}$$

$$\frac{d}{dx}(e^U) = e^U \frac{dU}{dx} \qquad \frac{d}{dx}(\cos U) = -\sin U \frac{dU}{dx}$$

C.5 Indefinite Integrals

If $U = U(x)$, $V = V(x)$, and $a = \text{constant}$,

$$\int a \, dx = ax + C$$

$$\int U^n \, dU = \frac{U^{n+1}}{n+1} + C, \quad n \neq -1$$

$$\int \frac{dU}{U} = \ln U + C \qquad \int e^{ax} \, dx = \frac{1}{a} e^{ax} + C$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax + C$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax + C$$

First-Order Differential Equation (I)

- A **separable equation** (可分离变量方程) is a first-order differential equations in which the expression for dy/dx can be factored as a function of x times a function of y . In other words, it can be written in the form $dy/dx=g(x)f(y)$. And if $f(y)\neq 0$, we could write $dy/dx=g(x)/h(y)$ where $h(y)=1/f(y)$
- A separable equation can hence be solved readily as shown below

$$\frac{dy}{dx} = \frac{g(x)}{h(y)} \rightarrow h(y) dy = g(x) dx \rightarrow \int h(y) dy = \int g(x) dx$$

- **Example:** Solve the differential equation $dy/dx=x^2/y^2$

$$\begin{aligned} \frac{dy}{dx} &= \frac{x^2}{y^2} \rightarrow y^2 dy = x^2 dx \\ \rightarrow \int y^2 dy &= \int x^2 dx \rightarrow \frac{1}{3} y^3 = \frac{1}{3} x^3 + C \rightarrow y = \sqrt[3]{x^3 + 3C} \end{aligned}$$

First-Order Differential Equation (II)

- The generic first-order linear differential equation has the form of

$$\frac{dy}{dx} + P(x)y = Q(x)$$

- Where P and Q are continuous functions on a given interval
- If we multiply both sides of the above equation by a suitable function $I(x)$, called **integrating factor** (积分因子), such that

$$I(x) \left(\frac{dy}{dx} + P(x)y \right) = (I(x)y)'$$

- Then the generic first-order differential equation becomes

$$(I(x)y)' = I(x)Q(x) \quad \Rightarrow \quad y = \frac{1}{I(x)} \left[\int I(x)Q(x)dx + C \right]$$

- A particular such integrating factor is given by (which can be easily confirmed)

$$I(x) = e^{\int P(x)dx}$$

Example: First-Order Differential Equation

- For the circuit shown right, find the current $i(t)$ if the switch is turned on at time $t = 0$
- Applying KVL, we get

$$iR + L \frac{di}{dt} = E(t)$$

- When $t \geq 0$, with the given numbers and we will achieve

$$12i + 4 \frac{di}{dt} = 60 \rightarrow \frac{di}{dt} + 3i = 15$$

- Thus, compared with the generic first-order differential equation, the functions are $P=3$ and $Q=15$. The integrating factor can then be

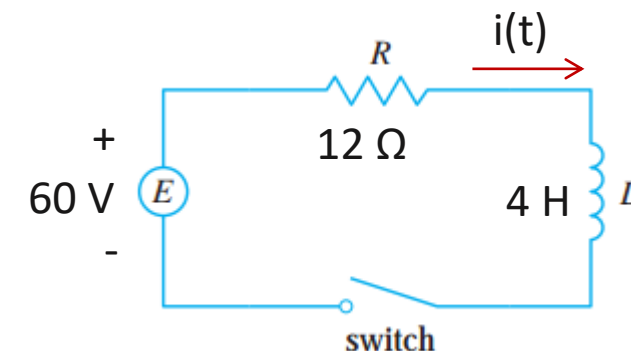
$$I(x) = e^{\int 3dt} = e^{3t}$$

- We can get

$$e^{3t}i = \int 15e^{3t}dt = 5e^{3t} + C \rightarrow i = 5 + Ce^{-3t}$$

- Since $i(0) = 0$ (the switch is open when $t < 0$), we have $C = -5$, and the solution is

$$i(t) = 5(1 - e^{-3t})$$



$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$y = \frac{1}{I(x)} \left[\int I(x)Q(x)dx + C \right]$$

$$I(x) = e^{\int P(x)dx}$$

Second-Order Differential Equation (I)

- Now we consider second-order differential equation. A second-order linear differential equation has the generic form of

$$P(x) \frac{d^2 y}{dx^2} + Q(x) \frac{dy}{dx} + R(x) y = G(x)$$

- where P , Q , R , and G are continuous functions, and $P(x) \neq 0$
- We consider a second-order homogeneous differential equation first, i.e., $G(x)=0$, so

$$P(x) \frac{d^2 y}{dx^2} + Q(x) \frac{dy}{dx} + R(x) y = 0$$

- The following two theorems will help us in solving the second-order homogeneous differential equation
 - If $y_1(x)$ and $y_2(x)$ are both solutions, then the function $y(x) = c_1 y_1(x) + c_2 y_2(x)$, where c_1 and c_2 are any constants, is also a solution.
 - If $y_1(x)$ and $y_2(x)$ are **linearly independent solutions** (线性无关解), i.e., none of them is a multiple of another, then the **general solution** (通解) is given by $y(x) = c_1 y_1(x) + c_2 y_2(x)$, where c_1 and c_2 are arbitrary constants

Second-Order Differential Equation (II)

- It is always possible to discover particular solutions (特解) to a second-order linear equation if the coefficient functions P, Q, and R are constant functions

$$ay'' + by' + cy = 0$$

- Where a, b, and c are constants and $a \neq 0$. We will see constantly that this is the case in this course, for the linear and time-invariant circuits
- If $y = e^{rx}$ is a solution, then we have

$$ar^2 e^{rx} + bre^{rx} + ce^{rx} = 0 \rightarrow (ar^2 + br + c)e^{rx} = 0 \rightarrow ar^2 + br + c = 0$$

- The equation $ar^2 + br + c = 0$ is called the **characteristic equation** (特征方程) of the differential equation $ay'' + by' + cy = 0$. We can use quadratic formula to find its roots

$$ar^2 + br + c = 0 \rightarrow r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

Second-Order Differential Equation (II)

- The equation $ar^2 + br + c = 0$ is called the **characteristic equation** (特征方程) of the differential equation $ay'' + by' + cy = 0$. We can use quadratic formula to find its roots

$$ar^2 + br + c = 0 \rightarrow r_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad r_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

- We distinguish three cases according to the sign of the discriminant $b^2 - 4ac$
 - Case I: $b^2 - 4ac > 0$, the roots r_1 and r_2 of the characteristic equation are real and distinct, and e^{r_1x} and e^{r_2x} are linearly independent, so the general solution is

$$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$$

- Case II: $b^2 - 4ac = 0$, the roots are real and equal, the general solution is

$$y = (c_1 + c_2 x) e^{rx}$$

- Case III: $b^2 - 4ac < 0$, the roots r_1 and r_2 of the characteristic equation are **complex numbers** (复数)

$$r_1 = \alpha + j\beta, \quad r_2 = \alpha - j\beta, \quad \text{where } \alpha = -\frac{b}{2a}, \quad \beta = \frac{\sqrt{|b^2 - 4ac|}}{2a}$$

Second-Order Differential Equation (IV)

- We distinguish three cases according to the sign of the discriminant $b^2 - 4ac$
 - Case III: $b^2 - 4ac < 0$, the roots r_1 and r_2 of the characteristic equation are complex numbers (复数)

$$r_1 = \alpha + j\beta, r_2 = \alpha - j\beta, \text{ where } \alpha = -\frac{b}{2a}, \beta = \frac{\sqrt{|b^2 - 4ac|}}{2a}$$

- With the help of Euler's formula (欧拉公式), i.e., $e^{jx} = \cos(x) + j \cdot \sin(x)$, we can achieve the general solution

$$y = C_1 e^{r_1 x} + C_2 e^{r_2 x} = e^{\alpha x} [(C_1 + C_2) \cos \beta x + j(C_1 - C_2) \sin \beta x] \quad \text{or} \quad \boxed{y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)}$$

- Summary: solution

Roots of $ar^2 + br + c = 0$	General solution
r_1, r_2 real and distinct	$y = c_1 e^{r_1 x} + c_2 e^{r_2 x}$
$r_1 = r_2 = r$	$y = c_1 e^{rx} + c_2 x e^{rx}$
r_1, r_2 complex: $\alpha \pm i\beta$	$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$

Example: Second-Order Differential Equation

- Solve the equations

$$(1) y'' + y' - 6y = 0; (2) 4y'' + 12y' + 9y = 0; (3) y'' - 6y' + 13y = 0.$$

- (1) The characteristic equation is

$$r^2 + r - 6 = 0 \rightarrow r_1 = 2, r_2 = -3$$

➤ The general solution is $y = c_1 e^{2x} + c_2 e^{-3x}$

- (2) The characteristic equation is

$$4r^2 + 12r + 9 = 0 \rightarrow r_1 = r_2 = -\frac{3}{2}$$

➤ The general solution is

$$y = c_1 e^{-3x/2} + c_2 x e^{-3x/2}$$

- The characteristic equation is

$$r^2 - 6r + 13 = 0 \rightarrow r_1 = 3 + 2j, r_2 = 3 - 2j$$

➤ The general solution is $y = e^{3x} (c_1 \cos 2x + c_2 \sin 2x)$

- Consider the second-order nonhomogeneous linear differential equations with constant coefficients, i.e., equations of the form

$$ay'' + by' + cy = G(x)$$

- The general solution of the nonhomogeneous differential equation can be written as

$$y(x) = y_p(x) + y_c(x)$$

- Where $y_p(x)$ is a **particular solution (特解)** of the nonhomogeneous differential equations $ay'' + by' + cy = G(x)$ while $y_c(x)$ is the general solution of the **complementary equation (补余方程)** $ay'' + by' + cy = 0$
- Will discuss more in the second-order circuits later